

Databehandling SOP - Forsøg 2

The optimal values are gives as:

$$> \alpha := 135$$

$$\alpha := 135 \quad (1)$$

$$> \alpha_p := 0$$

$$\alpha_p := 0 \quad (2)$$

$$> \beta := 157.5$$

$$\beta := 157.5 \quad (3)$$

$$> \beta_p := 22.5$$

$$\beta_p := 22.5 \quad (4)$$

Angles and corrsponding coincidences

$$> N(0, 22.5) := 1190$$

$$N(0, 22.5) := 1190 \quad (1.1)$$

$$> N(0, 67.5) := 245$$

$$N(0, 67.5) := 245 \quad (1.2)$$

$$> N(0, 112.5) := 137$$

$$N(0, 112.5) := 137 \quad (1.3)$$

$$> N(0, 157.5) := 1263$$

$$N(0, 157.5) := 1263 \quad (1.4)$$

$$> N(45, 22.5) := 429$$

$$N(45, 22.5) := 429 \quad (1.5)$$

$$> N(45, 67.5) := 132$$

$$N(45, 67.5) := 132 \quad (1.6)$$

$$> N(45, 112.5) := 706$$

$$N(45, 112.5) := 706 \quad (1.7)$$

$$> N(45, 157.5) := 1034$$

$$N(45, 157.5) := 1034 \quad (1.8)$$

$$> N(90, 22.5) := 114$$

$$N(90, 22.5) := 114 \quad (1.9)$$

$$> N(90, 67.5) := 364$$

$$N(90, 67.5) := 364 \quad (1.10)$$

$$> N(90, 112.5) := 398$$

$$N(90, 112.5) := 398 \quad (1.11)$$

$$\begin{aligned} & \text{> } N(90, 157.5) := 114 \\ & \text{ } N(90, 157.5) := 114 \end{aligned} \quad (1.12)$$

$$\begin{aligned} & \text{> } N(135, 22.5) := 891 \\ & \text{ } N(135, 22.5) := 891 \end{aligned} \quad (1.13)$$

$$\begin{aligned} & \text{> } N(135, 67.5) := 589 \\ & \text{ } N(135, 67.5) := 589 \end{aligned} \quad (1.14)$$

$$\begin{aligned} & \text{> } N(135, 112.5) := 91 \\ & \text{ } N(135, 112.5) := 91 \end{aligned} \quad (1.15)$$

$$\begin{aligned} & \text{> } N(135, 157.5) := 346 \\ & \text{ } N(135, 157.5) := 346 \end{aligned} \quad (1.16)$$

The expectation value is given as following

$$E(a, b) = \frac{N(a, b) + N(a^\perp, b^\perp) - N(a^\perp, b) - N(a, b^\perp)}{N(a, b) + N(a^\perp, b^\perp) + N(a^\perp, b) + N(a, b^\perp)}$$

The expectation value is dependent on the correlation function $N(\alpha, \beta)$, and hereby the perpendicular values of a, b written as $a^\perp, b^\perp$. A function for the perpendicular value is therefor defined as

$perp(v)$

$$\begin{aligned} & \text{> } perp(v) := \begin{cases} v - 90 & v + 90 > 180 \\ v + 90 & v - 90 < 0 \end{cases} \\ & \text{ } perp := v \mapsto \begin{cases} v - 90 & 0 < v - 90 \\ v + 90 & v < 90 \end{cases} \end{aligned} \quad (5)$$

$$\begin{aligned} & \text{> } E(a, b) := \frac{N(a, b) + N(perp(a), perp(b)) - N(perp(a), b) - N(a, perp(b))}{N(a, b) + N(perp(a), perp(b)) + N(perp(a), b) + N(a, perp(b))} \\ & \text{ } E := (a, b) \mapsto \frac{N(a, b) + N(perp(a), perp(b)) - N(perp(a), b) - N(a, perp(b))}{N(a, b) + N(perp(a), perp(b)) + N(perp(a), b) + N(a, perp(b))} \end{aligned} \quad (6)$$

The given expectation values has to be defined for the gives S value

$$S = E(\alpha, \beta) - E(\alpha, \beta_p) + E(\alpha_p, \beta) + E(\alpha_p, \beta_p)$$

$E(\alpha, \beta)$

Expectation value for the given angles:

$$\begin{aligned} & \text{> } a := \alpha \\ & \text{ } a := 135 \end{aligned} \quad (2.1)$$

$$\begin{aligned} & \text{> } b := \beta \\ & \text{ } b := 157.5 \end{aligned} \quad (2.2)$$

$$\begin{aligned} & \text{> } E(a, b) \\ & \end{aligned} \quad (2.3)$$

	$-\frac{1145}{2101}$	(2.3)
$\begin{array}{l} \text{> } E_{\alpha, \beta} := E(a, b) \end{array}$		
	$E_{\alpha, \beta} := -\frac{1145}{2101}$	(2.4)
$\begin{array}{l} \text{> } evalf(\%) \end{array}$	-0.5449785816	(2.5)

$E(\alpha, \beta_p)$		
$\begin{array}{l} \text{> } a := \alpha \end{array}$	$a := 135$	(3.1)

$\begin{array}{l} \text{> } b := \beta_p \end{array}$	$b := 22.5$	(3.2)
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$\begin{array}{l} \text{> } E(a, b) \end{array}$	$\frac{1077}{2117}$	(3.3)
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$\begin{array}{l} \text{> } E_{\alpha, \beta_p} := E(a, b) \end{array}$	$E_{\alpha, \beta_p} := \frac{1077}{2117}$	(3.4)
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$\begin{array}{l} \text{> } evalf(\%) \end{array}$	0.5087387813	(3.5)
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$E(\alpha_p, \beta)$		
$\begin{array}{l} \text{> } a := \alpha_p \end{array}$	$a := 0$	(4.1)

$\begin{array}{l} \text{> } b := \beta \end{array}$	$b := 157.5$	(4.2)
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$\begin{array}{l} \text{> } E(a, b) \end{array}$	$\frac{634}{993}$	(4.3)
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$\begin{array}{l} \text{> } E_{\alpha_p, \beta} := E(a, b) \end{array}$	$E_{\alpha_p, \beta} := \frac{634}{993}$	(4.4)
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$\begin{array}{l} \text{> } evalf(\%) \end{array}$		
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[		0.6384692850	(4.5)
	$E(\alpha_p, \beta_p)$		
[	> $a := \alpha_p$		
		$a := 0$	(5.1)
[	> $b := \beta_p$		
		$b := 22.5$	(5.2)
[	> $E(a, b)$		
		$\frac{1337}{1839}$	(5.3)
[	> $E_{\alpha_p, \beta_p} := E(a, b)$		
		$E_{\alpha_p, \beta_{-p}} := \frac{1337}{1839}$	(5.4)
[	> $evalf(\%)$		
		0.7270255574	(5.5)
[	Following the formula, the S value is defined by:		
[	> $S := E_{\alpha\beta} - E_{\alpha\beta_p} + E_{\alpha_p\beta} + E_{\alpha_p\beta_{-p}}$		
		$S := \frac{281371509445}{902475412751}$	(7)
[	> $evalf(S)$		
		0.3117774794	(8)
[	Although an alternative formulation for S_1 gives a more expected S value:		
[	> $S_1 := -E_{\alpha\beta} + E_{\alpha\beta_p} + E_{\alpha_p\beta} + E_{\alpha_p\beta_{-p}}$		
		$S_1 := \frac{2183279533497}{902475412751}$	(9)
[	> $evalf(S_1)$		
		2.419212205	(10)
[	The QU-tools resultats are as following:		
	Integration Time: 1000 ms		
	CHSH Result: 2.416		
	CHSH Error: 0.035		